

Investigating Generative AI's Impact on Cognitive Effort and Long-Term Memory in
Adolescents: A Mixed-Methods Study

April 30, 2025

Word Count: 4808

Abstract

Despite the growing prevalence of generative AI in education, little research has been done on how its use may affect students' long-term memory due to a reduced cognitive effort to process simplified second hand AI information. This study looks at whether increased reliance on generative AI tools for learning correlates with reduced cognitive engagement and lower long-term retention of academic material among adolescent students. A mixed-methods study was utilized. Part 1 involved a survey of 67 high school students assessing AI usage frequency, study habits, and perceived effectiveness through a qualitative approach. Part 2 employed a quantitative approach, including an experimental assessment with 10 participants who engaged with two passages, one with AI assistance and one without, followed by a test two weeks later to measure retention. Qualitative results showed students generally perceived AI tools as helpful for understanding but expressed uncertainty when asked about its impact on long-term retention. Quantitative data revealed minimal differences in test performance between AI and non-AI groups, though in nonfiction comprehension when AI was used, scores had a slight positive shift. These findings suggest that while AI tools enhance immediate comprehension, their influence on deeper cognitive processing and memory retention may be more limited, raising the question as to whether or not the trade-off between convenience and depth for learning will be beneficial for an AI-integrated society.

Introduction

AI Today

Over the last few years, learning methods have also undergone tremendous changes, primarily through increased access to AI technologies. As AI tools become more available across the world, they are integrating into daily life and changing the way ordinary information is processed (Zhang et. al., 2024). Whether that's asking language models, such as Open AI's ChatGPT for a more simplified read of a local news article, or using visual processing AI tools such as PhotoMath to break down the steps for a complex math problem, locating desired information has simplified significantly. The most commonly used AI by the general public is Generative AI, which according to Belkina et. al. (2025) refers to technology that employs deep learning models to create content resembling human output, such as images or text, in response to a wide range of inputs like questions, instructions, or language-based prompts. Such examples include OpenAI's ChatGPT, Microsoft Copilot, and Google Gemini, which have all substantially increased in global use since November 2022 (Zhang et. al., 2024).

AI Usage by Students

Looking at current statistics, more than 50% of teens from a survey sample of 1500 report that they use AI for school related work to get quick information and brainstorm ideas (Nagelhout, 2024). Especially for Gen Z adolescent learners who grew up amidst this rapid shift, it is crucial to assess the efficacy of AI as a learning tool for school, higher education, and careers, in order to gauge the level of depth they truly have on foundational learning subjects, and how they will use that to lead the next generation of learners. Students are finding benefits to

their learning ability by using AI. For instance, it allows them to explore new ideas, unlocking more creativity, and can even mimic an external tutor when they need to dig deeper on questioning and understanding difficult concepts (Nagelhout, 2024). With personalized instruction, automatic feedback, and learning support even outside of the classroom, AI for education seems like an obvious positive factor because it improves student efficiency to learn with more aid. Yet, with AI developments being a recent innovation, there is not much awareness as to some of the cons with AI aided learning. Most popular discussion regarding negative AI factors talk about ethics, with educators claiming students can use it for academic dishonesty, to AI developers even confessing that their models could have biased data and will not have 100% accuracy (Khan et. al., 2023). However, rather than looking at the external consequences of AI use, there has not been much focus intrinsically. And circling back to the fact that 50% of teens use AI for school related work (Nagelhout, 2024), it shows that in 2 years since its mass debut, AI is already becoming a dependent resource.

AI Overdependence

This raises the concept of AI over-dependency, or utilizing AI tools too frequently to learn, progress, and obtain information to the extent where independent cognitive thinking is reduced. In a study by Zhang et. al. on problematic AI usage behavior, they surveyed 300 university students and found that most reported effect of AI over dependency was reduced critical thinking ability, and increased spread of misinformation (2024). With AI overdependence being negative to critical thinking, there could be a deeper explanation at the neural level. AI is commonly used by students to answer a short-term problem quickly (Nagelhout, 2024). These

often result in quick, simple answers, to And according to Katharine Miller, the simpler a difficult task is made, the more likely a person is to continue using that AI to keep learning, creating an over reliance on it (2024). However, when this short term fix starts becoming a routine habit, it potentially could hinder long term memory of the topic. For instance, the oversimplified responses AI gives to make topics more digestible reduces the cognitive effort taken to comprehend the topic in the long term (Miller, 2023).

Processing Simple vs. Difficult Concepts

These are just a few of the conflicting perspectives on this issue. To better understand, the topics can be subdivided into smaller focus groups. One perspective on this issue includes an educational lens. Research suggests that AI's primary strength lies in its ability to cater its ability to the user, adapting to personalize each student's needs and providing targeted feedback. A specific instance can be illustrated in Kazmi's study, where she experimented to find the optimal balance between AI and instructor led learning. She deduced that AI tools specifically for her math class increased student engagement and helped students complete their assignments faster, highlighting that AI is successfully able to work one on one with students to boost their performance (Kazmi, 2022). Additionally, expanding to a broader perspective of equity, Baumer and Frueh emphasize the benefits of AI for neurodiverse learners, specifically how personalized AI tools adjust to various cognitive abilities, which otherwise may be difficult to accommodate for in the group classroom.

Impact on Long-Term Retention

However, Miller argues that AI overreliance could hinder cognitive engagement by simplifying complex tasks into surface level understanding, rather than deeply processed material. While AI speeds up the duration of a task, Miller raises concerns about whether or not students are able to understand the reasoning behind the solutions (Miller, 2023). Master et. al. support this hypothesis by adding that mental work is often reduced by AI, which is detrimental to long-term memory formation (2023). Their study found a positive correlation between cognitive effort, or “how hard one tries”, and working memory, the type of memory you access immediately after being told to memorize something, by measuring pupil dilations based on easy or difficult tasks (pupil dilations indicated increased effort, while reduction in size indicated less effort). This emphasis on effort as a measurable predictor of memory ties directly into concerns about how AI may affect students’ depth of processing when learning new material.

Gap in Research

While existing research highlights how generative AI can improve efficiency and even short-term comprehension, there is a notable gap in directly connecting AI use to its impact on long-term memory. Studies like Master et al. focus specifically on working memory, not long-term retention, and do not account for the role of cognitive effort in encoding information for future recall. Other studies emphasize performance or user satisfaction but stop short of exploring how overreliance on AI tools might affect the depth of learning over time. Moreover, no current research has examined this question specifically in adolescents, a population increasingly using AI in academic settings. Understanding this gap in the research is crucial moving forward to reconstruct strong educational systems that prepare students well for society

with the right assets equipped. If AI merely accelerates learning without improving retention or the depth of understanding, its role in education will need to be reevaluated.

This study investigates generative AI's impact on learning by looking for a correlation with cognitive engagement and long-term memory. Because there is more access to generative AI tools, cognitive effort is decreasing and overreliance is increasing. Thus, this research paper investigates if these factors combined result in reduced long-term retention when processing new information or tasks. The hypothesis is that adolescents who use generative AI tools to comprehend a chosen topic will perceive an improved immediate understanding, but will have decreased long-term retention due to reduced cognitive effort during initial learning.

References

- Almasri, F. (2024). Exploring the impact of artificial intelligence in teaching and learning of science: A systematic review of empirical research. *Research in Science Education*, 54(1), 977–997. <https://doi.org/10.1007/s11165-024-10176-3>
- Aquino, S. (2024, August 23). *AI for people with disabilities: Accessibility*. Aquino Archive.
- Answer Everything. (2014). *Before the internet, librarians would*. NPR.
- Baumer, N., & Frueh, J. (2021). *What is neurodiversity?* Harvard Health.
- Dai, Y. (2024). Integrating unplugged and plugged activities for holistic AI education: An embodied constructionist pedagogical approach. *Education and Information Technologies*. <https://doi.org/10.1007/s10639-024-13043-w>
- Deslauriers, L., McCarty, L. S., Miller, K., Callaghan, K., & Kestin, G. (2019, September 4). *Study shows that students learn more when taking part in classrooms that employ active-learning strategies*. Harvard Gazette.
- Henneborn, L. (2023, August 18). *Designing generative AI to work for people with disabilities*.
- Kazmi, A. (2024). *Can AI transform education?* Bill & Melinda Gates Foundation.
- Kamalov, F., Santandreu Calonge, D., & Gurrib, I. (2023). New era of artificial intelligence in education: Towards a sustainable multifaceted revolution. *Sustainability*, 15(16), 12451. <https://doi.org/10.3390/su151612451>
- Khan, B., Fatima, H., Qureshi, A., Kumar, S., Hanan, A., Hussain, J., & Abdullah, S. (2023). Drawbacks of Artificial Intelligence and Their Potential Solutions in the Healthcare

- Sector. Biomedical materials & devices (New York, N.Y.), 1–8. Advance online publication. <https://doi.org/10.1007/s44174-023-00063-2>
- Magni, F., Park, J., & Chao, M. M. (2024). Humans as creativity gatekeepers: Are we biased against AI creativity? *Journal of Business and Psychology*, 39(1), 643–656. <https://doi.org/10.1007/s10869-023-09910-x>
- Marshall, P. (2000). The Fisher King. In *The Official ACT Prep Guide 2019-2020*. ACT, Inc.
- Miller, K. (2023). *AI overreliance is a problem*. Stanford University Human-Centered Artificial Intelligence.
- Nagelhout, R. (2024, September 10). *Students are using AI already. Here's what they think adults should know*. Harvard Graduate School of Education.
- Niklas H. Weber, Michael E. Dreyer, Depressurization induced vapor bubble growth in liquid methane during microgravity, *Cryogenics*, Volume 134, 2023, 103716, ISSN 0011-2275, <https://doi.org/10.1016/j.cryogenics.2023.103716>
- Sengar, S.S., Hasan, A.B., Kumar, S. et al. Generative artificial intelligence: a systematic review and applications. *Multimed Tools Appl* (2024). <https://doi.org/10.1007/s11042-024-20016-1>
- Zhang, S., Zhao, X., Zhou, T., & Kim, J. H. (2024). Do you have AI dependency? The roles of academic self-efficacy, academic stress, and performance expectations on problematic AI usage behavior. *International Journal of Educational Technology in Higher Education*, 21(1), 34. <https://doi.org/10.1186/s41239-024-00281-0>